Highlights

From detection to action: closing the autonomic loop of a self-coordinated manufacturing line with rule-based and retrieval-augmented LLM planners

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- The 2020 autonomic architecture implemented detection only, with no planning or execution.
- Wiring legacy alarms straight to repair actions performs worse than doing nothing.
- A classical adaptive change test is a strong baseline for alarm triage.
- A frontier LLM with retrieved knowledge matches it, and wins on intermittent faults.
- As a veto over batched alarms, the frontier LLM is misled by a detector artefact.

From detection to action: closing the autonomic loop of a self-coordinated manufacturing line with rule-based and retrieval-augmented LLM planners

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Abstract

Autonomic architectures for Industry 4.0 promise self-configuring, self-healing and self-optimising production, but published systems often stop at detection. We revisit our own 2020 self-coordinated manufacturing architecture and its simulator, and find that it implements Monitor and Analyze but not Plan, Execute or a Knowledge base: detected conditions only produce a log message. We port the simulator to a headless, tested engine over the Bosch production-line dataset and close the loop with an action space built on the line's parallel twin stations, a symbolic validator, fault injection (step, intermittent and drift faults) and an explicit effect model. We compare rule-based planners with large-language-model (LLM) planners with and without retrieval-augmented operational knowledge, on a local 7B model (Qwen2.5) and a frontier model (Claude Fable 5.1), using a development trace for design and three held-out traces for evaluation. Closing the loop naively is harmful: wiring the original alarms to actions loses against doing nothing. The bottleneck is alarm triage, not planning. A classical adaptive change test recovers 52–54% of the delay caused by faults. With the 7B model, the retrieval-augmented planner recovers 20%; with the frontier model the same design recovers 58%, outperforming the best rule in 50 of 72 scenarios ($p = 0.003$) with a pooled gain that is not significant overall but is large on intermittent faults, which defeat every rule. Used instead as a veto over the original detector, the frontier model over-interprets an artefact of how that

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detector batches alarms. Model capability matters, but it cannot repair a poorly designed Analyze–Knowledge interface.

Keywords: Autonomic computing, MAPE-K, Industry 4.0, Large language models, Retrieval-augmented generation, Fault management

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1. Introduction

Autonomic computing proposes systems that manage themselves through a Monitor–Analyze–Plan–Execute loop over a shared Knowledge base (MAPE-K) [1, 2]. In manufacturing, this loop is attractive because production lines degrade continuously and human supervision does not scale: a self-healing line would notice a degrading station and reconfigure itself before the delay accumulates.

In 2020 we proposed an architecture for self-coordinated manufacturing processes with self-* properties and evaluated it with a simulator, SSACSimulator, over the Bosch production-line dataset [3, 4]. Revisiting that code six years later, we found that it implements the Monitor and Analyze phases only. When one of its three detectors fires, the simulator writes *"launching the autonomous cycle for reconfiguration"* to a log panel; no plan is produced and nothing is executed. The loop was announced, not closed.

This paper closes it, and asks what closing it actually requires. Two developments make the question timely. First, large language models (LLMs) are increasingly proposed as planners for industrial decision making, from the control of modular production systems [5] to advanced planning and scheduling [6]. Second, retrieval-augmented generation (RAG) [7] offers a natural candidate for the Knowledge component of MAPE-K: instead of a hand-built ontology, the planner retrieves the operational history it needs. Whether either helps, and against which baseline, is an empirical question.

Our contributions are:

1. A faithful, tested, headless port of the 2020 simulator, and an analysis of its Analyze layer on the published data: one of the three detectors is unreachable, one is exercised by 0.004% of the events, and the third fires routinely on a fault-free trace (Section 3).
2. The missing Plan and Execute phases: an action space based on the parallel "twin" stations of the Bosch line, a symbolic validator, fault injection with three fault shapes, and an explicit, parameterised effect model whose unverifiable assumptions are swept rather than fixed (Section 4).

3. A comparison of rule-based planners against LLM planners with and without retrieved knowledge, and of a hybrid in which a statistical test triages and the LLM vetoes, on a local 7B model and a frontier model, under a protocol that separates a development trace from three held-out traces (Sections 5–7).
4. Evidence that the bottleneck of closing the loop is alarm triage rather than planning; that LLM capability is decisive for retrieval-augmented triage; and that a capable model can be misled by an artefact of the detector it receives alarms from (Section 8).

All code, data-processing scripts, prompts, stored model responses and results are available (Section 9).

2. Background and related work

Autonomic computing and self-adaptive systems.. The MAPE-K reference model [1] structures a self-managing system as a feedback loop around a managed resource. Engineering such loops is the subject of the self-adaptive systems literature [2], which stresses that the Analyze phase must separate genuine change from noise before any adaptation is planned. LLMs have recently been examined as components of such loops: a state-of-the-art review and research roadmap organises their potential uses around the MAPE-K functions [8, 9]; LLM-based agents have been embedded in MAPE-K-based multi-agent systems [10]; and MAPER extends MAPE-K with an LLM-based reasoning component to handle unanticipated runtime situations [11]. Our study differs in placing an LLM inside the loop of an industrial system whose detector we inherit unchanged, and in measuring it against strong non-LLM baselines on held-out data.

Self- properties in manufacturing..* Our earlier work integrated multi-agent systems, service orientation and autonomic computing for Industry 4.0 process coordination [3, 4, 12]. The present paper re-examines that architecture empirically rather than proposing a new one.

Change detection.. Detecting a change in the rate of an event stream is a classical problem [13]. Our strongest rule-based baselines are simple instances: a Poisson test against a static base rate, and a test against the recent history of the same stream.

LLMs and retrieval in industrial decision making.. RAG conditions a language model on documents retrieved for each query [7]. In manufacturing, LLM agents have been combined with digital twins to plan and control modular production [5] and applied to advanced planning and scheduling [6], and a recent review of LLMs for fault diagnosis in smart manufacturing covers, among other roles, retrieval-augmented fusion of industrial knowledge [14]. Our design deliberately includes a strong statistical baseline and control rules that isolate what the LLM contributes beyond them.

3. The 2020 system revisited

3.1. Simulator and data

SSACSimulator is a Qt/C++ discrete-event simulator. A process model, derived from the Bosch Production Line Performance dataset [15], describes 52 stations on four sub-lines plus begin and end nodes, 285 transitions (84 of them main flows), a throughput time of 6 420 min and 1 183 165 cases. Four event traces of 100 000 transitions each replay recorded piece movements; they span 54 to 168 hours of production.

3.2. Headless port

We ported the simulation core (about 1 440 lines of C++ logic; the remaining ∼960 lines are graphical user interface) to Python. The port reproduces the original's behaviour as read from its source, including details that change results: thresholds computed in single precision, a 100 000-event loading limit, and a lenient rule for malformed rows. A fast path advances only the pieces that can leave the system; the original per-tick loop is kept as a test oracle, and a regression test checks that both produce identical detections, decisions and piece lifetimes. A run takes 0.2–0.4 s. The test suite has 87 tests.

3.3. What the Analyze layer actually does

The original implements three detectors: a station whose mean transition time exceeds 1.3 times its reference; more than 10% of stations flagged within one batch of simultaneous events; and a mean piece lifetime exceeding 1.1 times the throughput time. On the first trace, only the second fires (39 alarms). The predicted-failure detector depends on a classifier label that is positive in 15 of 400 000 events across all traces (0.004%). The global detector is unreachable: the mean piece lifetime (636 min) is an order of magnitude below its threshold (7 062 min). Finally, 45 of the 285 transitions have a reference time of zero, so any positive duration flags their station.

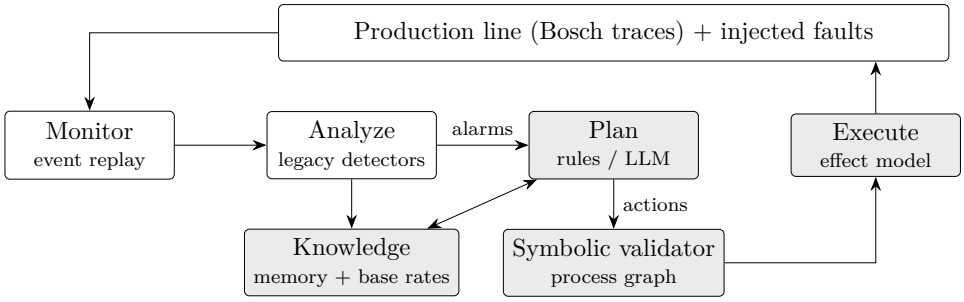


Figure 1: The closed MAPE-K loop. White: present in the 2020 system. Grey: added in this work.

Plan, Execute and Knowledge are absent. This is the starting point for the rest of the paper.

4. Closing the loop

Figure 1 summarises the closed loop. Components added in this work are shaded.

4.1. Twin stations and the action space

The Bosch line is built from parallel station pairs (and a few triples) that share predecessors and successors and split the load almost evenly; for example, stations L0_S2 and L0_S3 process 339 345 and 334 054 cases. We identify twins automatically from the overlap of their predecessor and successor sets on main flows: 29 of the 52 stations have one. Diverting a piece from a station to its twin keeps its trajectory consistent, which makes rerouting a physically meaningful action. The action space is Reroute(S, T, f), which diverts a fraction f of the pieces entering S to its twin T; Restore(S); Maintenance(S, d), which repairs S for d minutes; and NoOp.

4.2. Symbolic validator

Every action, from any planner, is checked against the process graph and the current interventions before execution: stations must exist, a reroute target must be a structural twin, reroutes must not form chains, and a station cannot be repaired twice concurrently. The validator never judges whether an action is a good idea, only whether it is admissible.

4.3. Effect model

The simulation is trace-driven: start times come from the recorded data and are not shifted; faults and actions act on transition durations, and the lifetime of a piece is the sum of its transition durations. Delay therefore does not propagate as queueing; the model estimates time lost per piece. A Bosch "transition" is the time between two station timestamps and mostly measures waiting: its median is 0 min and its 90th percentile 30 min, but its 99th percentile reaches 4 000–6 000 min. Effects are therefore *additive* and proportional to each station's processing time τ_S (the median of its transitions of up to 60 min), never multiplicative over a transition that includes days of storage. τ_S is 6–12 min and very stable across traces. The effect model is:

- **Fault** of factor k on S: $+(k(t) - 1)\,\tau_S$ on every transition leaving S, where $k(t)$ is a *step* (k from t_0), *intermittent* (k for 60 min, 1 for 120 min, repeating) or *drift* (rising linearly from 1 to k over 600 min).
- **Reroute**: a deterministic fraction of pieces (selected by a hash of the case identifier) goes to T, which then adds $\alpha\, f\, \tau_T$ for the extra load.
- **Maintenance**: faults on S that began before the end of the repair are cleared at its end. During the repair, in *contingency* mode (default) backup capacity outside the model processes the pieces, adding $0.5\,\tau_S$; in *stop* mode the pieces wait for the repair to finish. Repairing a healthy station costs and fixes nothing, which is what makes false alarms expensive.

The dataset is anonymised and provides no capacities, repair times or labelled failures, so all faults are synthetic and every parameter we could not ground in data (severity k, repair time d, congestion α, maintenance mode) is varied in a sensitivity analysis (Section 7.1).

5. Planners

All planners receive, for each alarm, the alarm and the state of their own past interventions. None sees the injected faults. All of them respond to a station judged faulty in the same way: reroute to a free twin if one exists, then repair.

5.1. Rule-based planners

- **Naive**: acts on every station named in an alarm.
- **Calibrated**: acts when the number of alarms on a station within 120 min is improbable (Poisson tail, $\alpha = 10^{-3}$) under its base rate, learned on a *different* fault-free trace.
- **Adaptive**: the same test, against the rate observed in the preceding 24 h of the same trace, causally, after an 8 h warm-up.
- **Adaptive + load veto** (control): the adaptive rule, ignoring stations that receive load the controller itself rerouted onto them.
- **Twin-only** (control): acts on every flagged station that has a free twin and ignores the others.

The two controls are not proposed methods; they implement as plain rules behaviours observed in LLM planners, so that an LLM's contribution is measured beyond them.

5.2. LLM planners

For every alarm with actionable stations, the LLM must return, for each of them, a verdict (*degraded*, *normal*, *self-inflicted* or *insufficient evidence*) followed by an action type (none, repair, or reroute and repair). The output is constrained by a JSON schema whose station names are restricted to the actionable ones, so the decoder cannot invent a station. Code chooses the twin, discards any action inconsistent with the model's own verdict, and passes the rest to the validator.

- **C1** receives the alarm, the actionable stations, whether each has a free twin, and the current interventions: the same structural information as the rules.
- **C2 (RAG)** additionally receives knowledge retrieved from the controller's operational memory: for each station, the alarms observed in the last 2 and 24 h against the number its fault-free base rate predicts, whether the controller is pushing rerouted load onto it, whether its alarms dropped after past repairs, and its most recent log records. Retrieval is keyed by station and ordered by recency; all statistics are computed by code.
- **Hybrid**: the adaptive change test selects the stations; the LLM, with C2's knowledge, confirms or vetoes and chooses the action. A vetoed station is not re-examined for 120 min.

Models.. We use Qwen2.5-7B-Instruct [16], 4-bit quantised, served locally with Ollama [17] at temperature 0; and Claude Fable 5.1 (Anthropic), accessed through the Claude Agent SDK [18] with effort set to *low*, no tools and no external context sources. The frontier model does not expose a temperature setting, so its runs are not bit-reproducible; every response of both models is stored, which makes all experiments replayable.

6. Experimental protocol

Scenarios.. A fault of factor $k = 3$ is injected 600 min after the start of a trace on one of eight stations, four with a twin (L0_S2, L0_S4, L0_S6, L3_S35) and four without (L0_S1, L0_S8, L3_S29, L3_S33), all of which have observable processing times on every trace, with one of the three fault shapes. Each trace also has a fault-free run. Repairs last 60 min, $\alpha = 0.5$, and maintenance runs in contingency mode unless stated otherwise.

Development and held-out traces.. Prompts and retrieved context were designed and revised only on the first trace, on four development cases; every version and its results were logged. The three other traces are held out: on them the prompts were run once, frozen. Calibrated rules always learn their base rates on a different trace from the one they are evaluated on.

Metrics.. Every run replays the same trace, so each piece is compared with itself in a fault-free reference run. The *excess delay* of a run is the sum over pieces of their extra lifetime. The *pooled recovery* of a planner over a set of scenarios is $1 - \sum \text{excess(planner)} / \sum \text{excess(no action)}$: 1 erases the cost of the faults; 0 is no better than doing nothing; negative values mean that the planner's own actions cost more than the faults. The *false-alarm cost* is the excess delay per hour in fault-free runs. A *specific hit* counts an action on the faulty station only if the same planner, on the same trace without the fault, did not act on that station within ± 120 min; it excludes routine repairs that happen to coincide with the fault.

Statistics.. Planners are compared on paired scenarios with the Wilcoxon signed-rank test on excess delay [19], with Holm adjustment [20], and with 95% bootstrap intervals [21] for the difference in pooled recovery, resampling scenarios within their trace (10 000 resamples).

Table 1: Rule-based planners on all four traces: pooled recovery (scenarios with net benefit). Central point: step, intermittent and drift faults, $k = 3$, 60 min repairs, $\alpha = 0.5$, contingency mode. Sensitivity rows: step faults, one parameter changed.

	Naive	Calibrated	Adaptive
All fault shapes	−0.11 (37/96)	+0.49 (73/96)	+0.56 (69/96)
step	+0.14 (17/32)	+0.59 (30/32)	+0.64 (29/32)
intermittent	−1.40 (3/32)	−0.09 (13/32)	+0.07 (13/32)
drift	+0.10 (17/32)	+0.60 (30/32)	+0.66 (27/32)
Step, $\alpha = 0$	–	+0.74	+0.73
Step, $\alpha = 1$	–	+0.47	+0.56
Step, $k = 1.5$	–	−0.39	+0.10
Step, $k = 5$	–	+0.75	+0.73
Step, repair 15 min	–	+0.84	+0.78
Step, repair 240 min	–	−0.80	+0.12
Step, stop mode	–	−1.17	−0.27

7. Results

7.1. Rule-based planners and sensitivity

Table 1 reports the rule-based planners over all four traces (96 fault scenarios; the rules have no trace-specific tuning) and a one-at-a-time sensitivity analysis on step faults. Wiring the original alarms directly to actions loses against doing nothing (−0.11): the naive planner repairs 122 healthy stations per scenario on average. The calibrated and adaptive rules recover about half of the fault cost. The result is robust to the congestion parameter α, but reverses for mild faults ($k = 1.5$), long repairs (240 min) and stop-mode maintenance, where the cost of false repairs outweighs the benefit; the adaptive rule degrades least. Intermittent faults defeat every rule.

7.2. Held-out comparison

Table 2 and Figure 2 report all planners on the three held-out traces (72 fault scenarios).

With the 7B model, both LLM planners that triage alarms themselves fall far below the rules. C1 behaves like the twin-only heuristic (on the development trace it made the same number of repairs as the twin-only control, with recoveries of +0.80 and +0.78 on one case and −0.16 and −0.19 on another), and retrieved knowledge improves it only modestly. Restricting the LLM to vetoing the adaptive rule's candidates (hybrid) reduces the false-alarm cost of the best rule by about 40% but loses detections.

Table 2: Held-out traces (72 fault scenarios, 3 fault-free runs). Recovery: pooled, with 95% stratified bootstrap interval. False repairs: repairs of healthy stations per fault scenario. False-alarm cost: piece-min/h in fault-free runs, mean of the three traces.

Planner	Model	Recovery	Specific hits	False repairs	False-alarm cost
Calibrated rule	–	+0.45 [0.37, 0.51]	32/72	48.8	108
Adaptive rule	–	+0.52 [0.43, 0.59]	57/72	32.8	67
Adaptive + load veto	–	+0.54 [0.45, 0.61]	57/72	31.2	64
Twin-only (control)	–	−0.31 [−0.41, −0.20]	9/72	65.0	142
C1	Qwen 7B	+0.15 [0.03, 0.26]	22/72	61.6	136
C2 (RAG)	Qwen 7B	+0.20 [0.07, 0.30]	31/72	74.0	159
Hybrid	Qwen 7B	+0.41 [0.31, 0.51]	50/72	24.0	38
C2 (RAG)	Fable 5.1	**+0.58** [0.51, 0.64]	39/72	32.8	58
Hybrid	Fable 5.1	+0.24 [0.13, 0.35]	34/72	**5.1**	**7**

With the frontier model, the ranking changes. The same C2 design reaches +0.58, the highest pooled recovery of all planners. Figure 3 places every planner on the trade-off between recovery under faults and the cost of acting on noise.

7.3. Statistical comparison

Table 3 compares C2 on Fable 5.1 with the other planners. It loses less production time than the best rule in 50 of 72 scenarios ($p = 0.003$, Holm-adjusted), but the difference in pooled recovery, +0.04, has a confidence interval that includes zero: it wins often and by little, and occasionally loses by more. The breakdown by fault shape locates the advantage (Figure 4): on step and drift faults the two planners are indistinguishable; on intermittent faults C2 on Fable 5.1 is better in 21 of 24 scenarios ($p = 0.005$, Holm-adjusted). The two planners on Fable 5.1 are the only ones with a positive pooled recovery on intermittent faults. In scenarios where both act on the faulty station, there is no evidence that it acts sooner (median difference 0 min, $p = 0.17$).

7.4. The hybrid and the batched-alarm artefact

The hybrid behaves in the opposite way with the frontier model: it becomes extremely conservative (5.1 false repairs per scenario, false-alarm cost 7 piece-min/h) and misses half of the faults. Its stated reasons explain why. Fable 5.1 vetoed at least one station in 1 256 of its 1 437 calls, and 98% of those vetoes argue that the alarms come in bursts shared by many stations and therefore point to a common, line-wide cause; all 167 vetoes of a station that was actually faulty give this reason. For example, with L0_S4 faulty: *"Every alarm is a line-wide burst (all L0 stations plus L3) with twins*

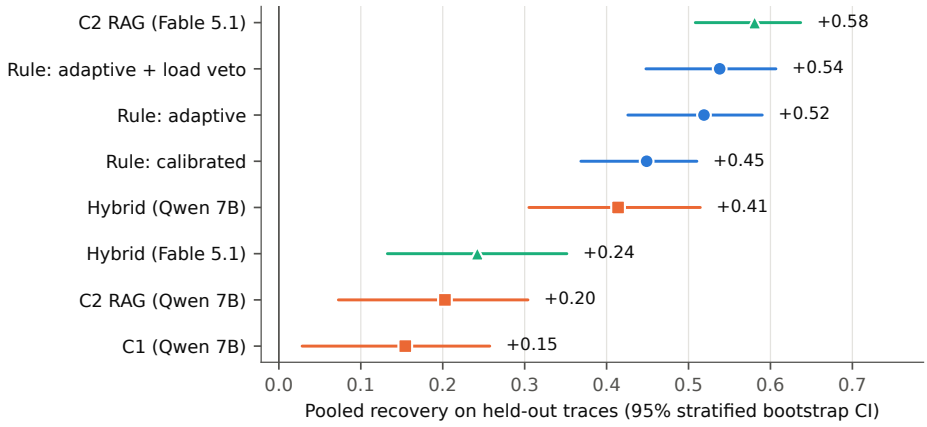


Figure 2: Pooled recovery on the held-out traces. Circles: rule-based planners; squares: Qwen2.5 7B; triangles: Claude Fable 5.1.

alarming together; points to a common upstream cause, not individual faults. Hold off." The inference is sound; its premise is an artefact. The original detector raises an alarm *only* when more than 10% of the stations are flagged in the same batch, so every alarm it produces is, by construction, a shared burst. With the 7B model, 17% of the vetoes give this reason.

8. Discussion

The bottleneck is triage, not planning.. Once a station is correctly judged faulty, a simple rule (reroute to a free twin, repair) is enough. Every difference we measured between planners comes from deciding which alarms deserve action. Closing the loop on the 2020 system without addressing this would have made the line worse.

Capability matters where the LLM does the triage.. The same retrieval-augmented design goes from +0.20 to +0.58 when the 7B model is replaced by a frontier model. Its advantage over a classical change test is concentrated on intermittent faults, whose on–off pattern dilutes alarm rates but leaves a recognisable history of alarms and repairs. On step and drift faults, a well-designed statistical test is as good, cheaper and fully reproducible.

A capable model cannot repair a poor Analyze–Knowledge interface.. As a veto over the legacy detector, the frontier model was less useful than the 7B model, because it reasoned correctly about evidence whose structure was

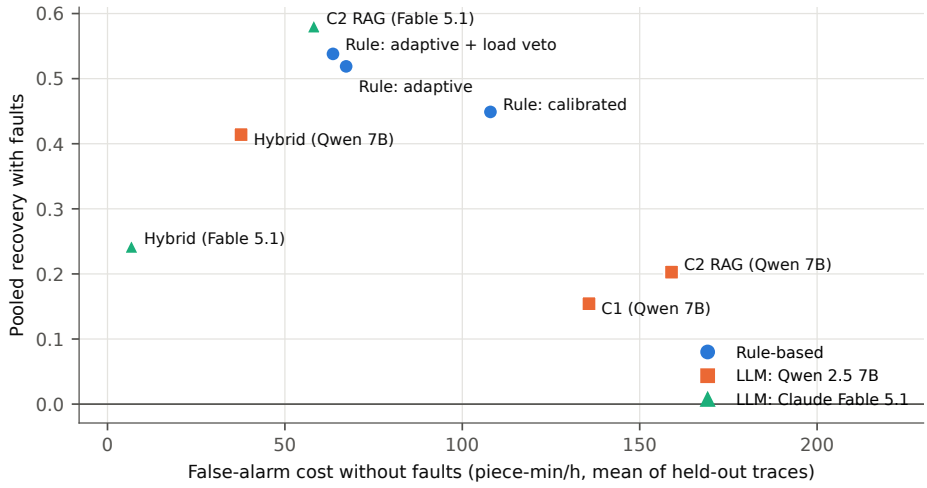


Figure 3: Recovery with faults against false-alarm cost without faults, held-out traces.

produced by the detector itself. In MAPE-K terms, the information passed from Analyze to Plan (batched alarms) discards exactly the per-station evidence the planner needs. The direct implication for the architecture is to redesign that interface, passing per-station evidence rather than batched alarms, before investing in more capable planners.

Development sets mislead.. On the development trace, three conclusions were wrong: the 7B retrieval-augmented planner appeared to beat the calibrated rule, the hybrid appeared to dominate its control, and the frontier model appeared equivalent to the 7B model. Only the held-out traces reversed them. A single trace with four cases is too small to select designs on, and LLM-based controllers should be evaluated on data not used to write their prompts.

Threats to validity.. *Construct*: faults are synthetic, and the effect model is additive and trace-driven, without queueing; its unverifiable parameters were swept, and the conclusions about triage hold across the swept range, but the absolute recoveries depend on them. Recovery is measured in production time only, without economic weights. *Internal*: the specific-hit metric depends on a ± 120 min window; the frontier model is not bit-reproducible, although every response is stored. *External*: one production line, one dataset and one legacy detector; two language models at the extremes of capability, with no intermediate sizes. *Statistical*: scenarios on

Table 3: C2 (RAG) on Claude Fable 5.1 against other planners, paired over held-out fault scenarios. Wilcoxon p-values Holm-adjusted within each block. Δ: difference in pooled recovery (positive favours C2 on Fable 5.1), 95% stratified bootstrap interval.

Against	Better in	p (Holm)	Δ recovery [95% CI]
Adaptive + load veto	50/72	0.003	+0.04 [−0.01, +0.10]
Adaptive rule	53/72	< 0.001	+0.06 [+0.01, +0.13]
Calibrated rule	63/72	< 0.001	+0.13 [+0.09, +0.18]
C2 (RAG), Qwen 7B	69/72	< 0.001	+0.38 [+0.31, +0.47]
Against adaptive + load veto, by fault shape			
step	14/24	0.46	+0.01 [−0.06, +0.10]
intermittent	21/24	0.005	+0.21 [0.00, +0.42]
drift	15/24	0.46	+0.02 [−0.06, +0.10]

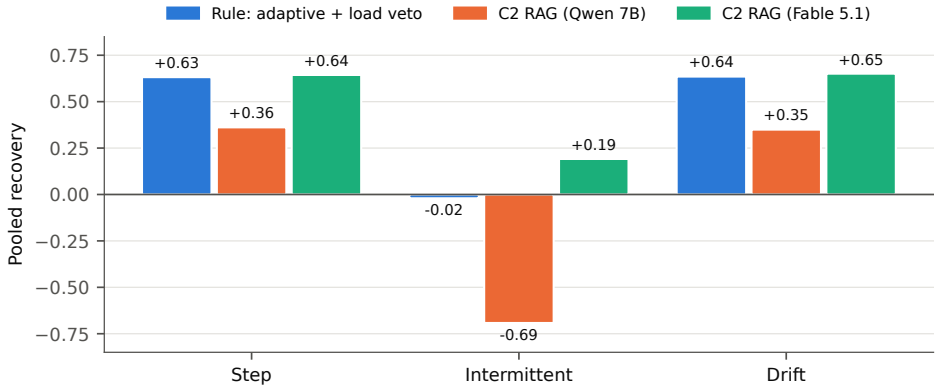


Figure 4: Pooled recovery by fault shape on the held-out traces.

the same trace are not independent; we resample within traces, but three held-out traces remain a small number of clusters.

9. Conclusion

We closed the autonomic loop of a self-coordinated manufacturing architecture that, as published, only detected. Doing so naively is harmful, because the decisive problem is alarm triage. A classical adaptive change test is a strong baseline. A frontier LLM with retrieved operational knowledge matches it overall and clearly outperforms it on intermittent faults, while the same design on a 7B model does not. Used as a veto over batched alarms, the frontier model is misled by the detector's own structure. Future

work will redesign the Analyze–Knowledge interface to pass per-station evidence, add intermediate model sizes, and validate the effect model against plant data with real capacities and repair times.

Data and code availability

The headless simulator, planners, prompts, development log, stored model responses, results and the scripts that produce every table and figure are available at [TODO: repository URL]. The Bosch dataset is publicly available [15].

CRediT authorship contribution statement

Manuel Sánchez: Conceptualization, Methodology, Investigation, Visualization, Writing – original draft. **Ernesto Exposito:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[TODO: Authors to complete and confirm. Draft: During the preparation of this work the authors used Claude (Anthropic) to assist with software development, data analysis and drafting of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. Note that the use of Qwen2.5 and Claude Fable 5.1 as planners is part of the method, not of manuscript preparation.]

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